Bochner formula & Laplacian comparison

(M,g) n-dim, $\quad Ric \geq (n-1)H \qquad H \in \mathbb{R}$

<u>Bochner</u>: $\quad \frac{1}{2}\Delta|\nabla u|^2 = |Hess_u|^2 + \langle \nabla u, \nabla \Delta u\rangle + Ric(\nabla u, \nabla u)$

Let $p \in M$, $\quad r : M \to \mathbb{R}$

$r(x) = d(p,x)$

wherever r is smooth (away from p and the cut locus C_p)

$|\nabla r| = 1$

$\nabla_{\nabla r}\nabla r = 0$

$\Longrightarrow$ $Hess_r$ has eigenvalues $\underset{\substack{\uparrow \\ \text{in direction} \\ \nabla r}}{0}, \lambda_1, \dots, \lambda_{n-1}$

Along minimizing geodesic from p

Let $m(r) = \Delta r$, so $m'(r) = \langle \nabla r, \nabla \Delta r\rangle$

$$\Longrightarrow \quad 0 = |Hess_r|^2 + m'(r) + Ric(\nabla r, \nabla r)$$
$$\geq |Hess_r|^2 + m'(r) + (n-1)H \qquad (*)$$

$$\left.\begin{aligned} \Delta r &= \lambda_1 + \dots + \lambda_{n-1} \\ |Hess_r|^2 &= \lambda_1^2 + \dots + \lambda_{n-1}^2 \end{aligned}\right\} \Longrightarrow \left(\frac{\Delta r}{n-1}\right)^2 \leq \frac{|Hess_r|^2}{n-1}$$

i.e. $\quad |Hess_r|^2 \geq \frac{m(r)^2}{n-1}$

$$(*) \Longrightarrow \quad m'(r) \leq -\frac{m(r)^2}{n-1} - (n-1)H$$

Let $\underline{m}(r)$ be the corresponding function in $\underline{M}^n_H$ = simply connected n-fold with curvature H

Then $$\underline{m}'(r) = \frac{-\underline{m}^2}{n-1} - (n-1)H$$

$$m(r), \ \underline{m}(r) = \frac{n-1}{r} + O(r) \quad \text{as } r \to 0.$$

In geodesic polar coords $g = dr^2 + g(\theta, r)$

$g(\theta, r)$ metric on S^{n-1}

Vol form is $dr\, \mathcal{A}(\theta, r)$.

along a given geodesic γ we write $\mathcal{A}(r)$ (θ determined by γ)

$m(r)$ = mean curvature of $\partial B(p, r)$

$$\Longrightarrow \quad m(r) = \frac{\mathcal{A}'(r)}{\mathcal{A}(r)}$$

Write $\underline{\mathcal{A}}$ for corresponding ~~function~~ vol. form on $\underline{M}^n_H$.

view $\mathcal{A}, \underline{\mathcal{A}}$ as functions relative to standard area form on S^{n-1}.

Define u, k by

$$\mathcal{A} = u^{n-1} \qquad \underline{\mathcal{A}} = k^{n-1}$$

$$m = (n-1)\frac{u'}{u} \qquad \underline{m} = (n-1)\frac{k'}{k}$$

$$u'' \leq -Hu \qquad k'' = -Hk$$

$$[uk(m-\underline{m})]' = [(n-1)(ku' - uk')]'$$
$$= (n-1)(ku'' - uk'')$$
$$\leq 0$$

$m - \underline{m} \to 0$ as $r \to 0$

$\Rightarrow \quad m - \underline{m} \leq 0$ as long as $uk > 0$ i.e. along a minimizing geodesic.

i.e. $m \leq \underline{m}$

$\Rightarrow \quad (\ln \mathcal{A})' \leq (\ln \underline{\mathcal{A}})'$

so $\dfrac{\mathcal{A}(r)}{\underline{\mathcal{A}}(r)}$ is decreasing (volume comparison)

<u>Laplacian comparison</u> (local)

$\Delta r = m$ at smooth points of r

$$\Rightarrow \quad \Delta f(r) = f''(r) + f'(r)\Delta r$$
$$= f''(r) + f'(r)\, m(r)$$

on $\underline{M}^n_H$ we have $\underline{\Delta} f(r) = f''(r) + f'(r)\,\underline{m}(r)$

$$\Longrightarrow \quad \Delta f(r) \leq \underline{\Delta} f(r) \quad \text{if} \quad f'(r) \geq 0$$
$$\Delta f(r) \geq \underline{\Delta} f(r) \quad \text{if} \quad f'(r) \leq 0$$

Can use these to construct useful barrier functions for use with the max principle.

Segment inequality

For a function $g \geq 0$ let

$$\mathcal{F}_g(x_1, x_2) = \inf_\gamma \int_0^{\ell(\gamma)} g(\gamma(s))\, ds$$

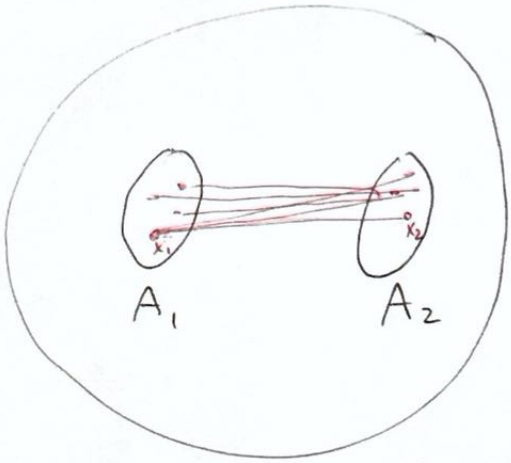
infimum is over all minimizing geodesics γ from x_1 to x_2, ds = arc length

Theorem (Cheeger-Colding) Let $Ric_{M^n} \geq -(n-1)$

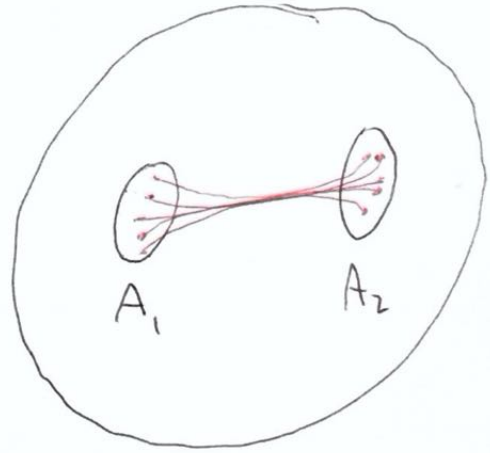
$A_1, A_2 \subset B(p, r)$, $R \geq r$

$$\int_{A_1 \times A_2} \mathcal{F}_g(x_1, x_2) \leq c(n, R)\, r \left(Vol(A_1) + Vol(A_2) \right) \int_{B_{2R}(p)} g$$

$$\frac{1}{2} c(n, R) = \frac{\mathcal{A}(2R)}{\mathcal{A}(R)} \quad (\text{in } \underline{M}^n_{-1})$$



$Ric \geq -(n-1)$



very negative curvature

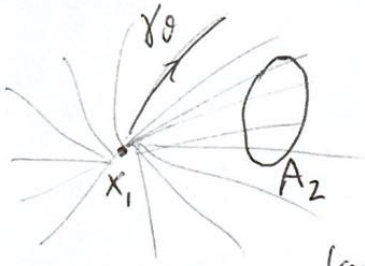
<u>Proof</u> Enough to work with $(x_1, x_2) \in A_1 \times A_2$ s.t. $\exists$ unique minimizing geodesic $\gamma_{x_1 x_2}$ from x_1 to x_2

Let
$$\mathcal{F}_{g,1}(x_1, x_2) = \int_{\frac{1}{2}d(x_1,x_2)}^{d(x_1,x_2)} g(\gamma_{x_1x_2}(s))\, ds$$

$$\mathcal{F}_{g,2} = \int_0^{\frac{1}{2}d(x_1,x_2)} g(\gamma_{x_1x_2}(s))\, ds$$

$$\mathcal{F}_g = \mathcal{F}_{g,1} + \mathcal{F}_{g,2}$$

Fix x_1.
$$\int_{A_2} \mathcal{F}_{g,1}(x_1, x_2) \le \int_{S^{n-1}} \int_0^{\ell(\theta)} \mathcal{F}_{g,1}(x_1, \gamma_\theta(u))\, du\, \mathcal{A}(\theta, u)$$



$\ell(\theta) \le 2r$
largest number up to where $\gamma_\theta(\ell(\theta))$ is minimizing

$$= \int_{S^{n-1}} \int_0^{\ell(\theta)} \int_{\frac{1}{2}u}^{u} g(\gamma_\theta(s))\, ds\, du\, \mathcal{A}(\theta, u)$$

$$\frac{\mathcal{A}(\theta, u)}{\mathcal{A}(\theta, s)} \le \frac{\mathcal{A}(u)}{\mathcal{A}(s)} \le \frac{1}{2} C(n, R)$$

$$\le \frac{1}{2} C(n, R) \int_{S^{n-1}} \int_0^{\ell(\theta)} \int_{\frac{1}{2}u}^{u} g(\gamma_\theta(s))\, ds\, \mathcal{A}(\theta, s)\, du$$

$$\leq \frac{1}{2} c(n,R) \int_0^{\ell(\theta)} \left(\int_{B_{2R}(p)} g \right) du$$

$$\leq c(n,R)\, r \int_{B_{2R}(p)} g$$

$$\Rightarrow \int_{A_1 \times A_2} F_{g,1}(x_1, x_2) \leq c(n,R)\, r\, \mathrm{Vol}(A_1) \int_{B_{2R}(p)} g$$

$\square$

Same holds for $F_{g,2}$ w/ A_1 & A_2 interchanged.